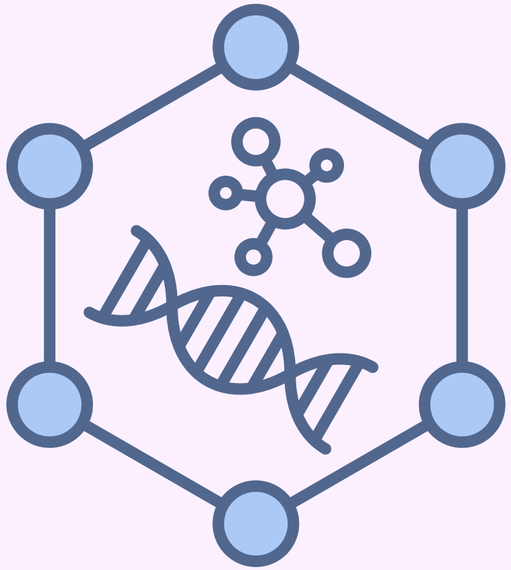

Title: Design and Analysis of Bio-Inspired Self-Healing for Next-Gen Orthopedic Implant

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Introduction

- Current orthopedic implants often suffer from "stress shielding" and fatigue.
- This project explores Biomimetic design, specifically looking at the trabecular structure of bone.
- Goal: Create a synthetic micro-lattice that mimic's bone's ability to redistribute load and "heal" via micro-encapsulated polymers.

Problem Statement

- Traditional metallic implants (Titanium/Stainless Steel) are too rigid compared to natural bone.
- Lack of "self-repair" leads to implant loosening and secondary surgeries.
- Need for a lightweight, porous structure that promotes osseointegration while maintaining high structural integrity.

Functional Requirements

- FR1: The structure must match the Young's Modulus of cortical and cancellous bone.
- FR2: Must provide a 70-80% porosity for nutrient flow and cell growth.
- FR3: Integration of micro-channels for the delivery of healing agents.

Non-functional Requirements

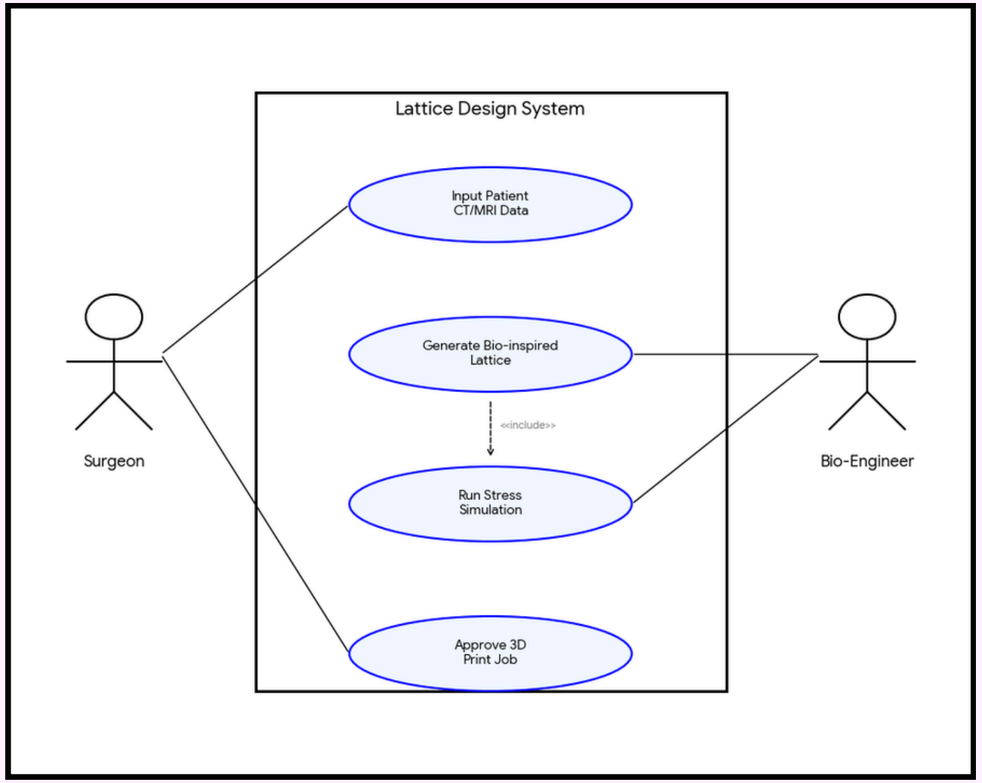
- Biocompatibility: Must not trigger an immune response.
- Durability: Must withstand 10^7 cycles of cyclic loading (gait simulation).
- Scalability: Design must be compatible with SLS (Selective Laser Sintering) 3D printing.

System Architecture

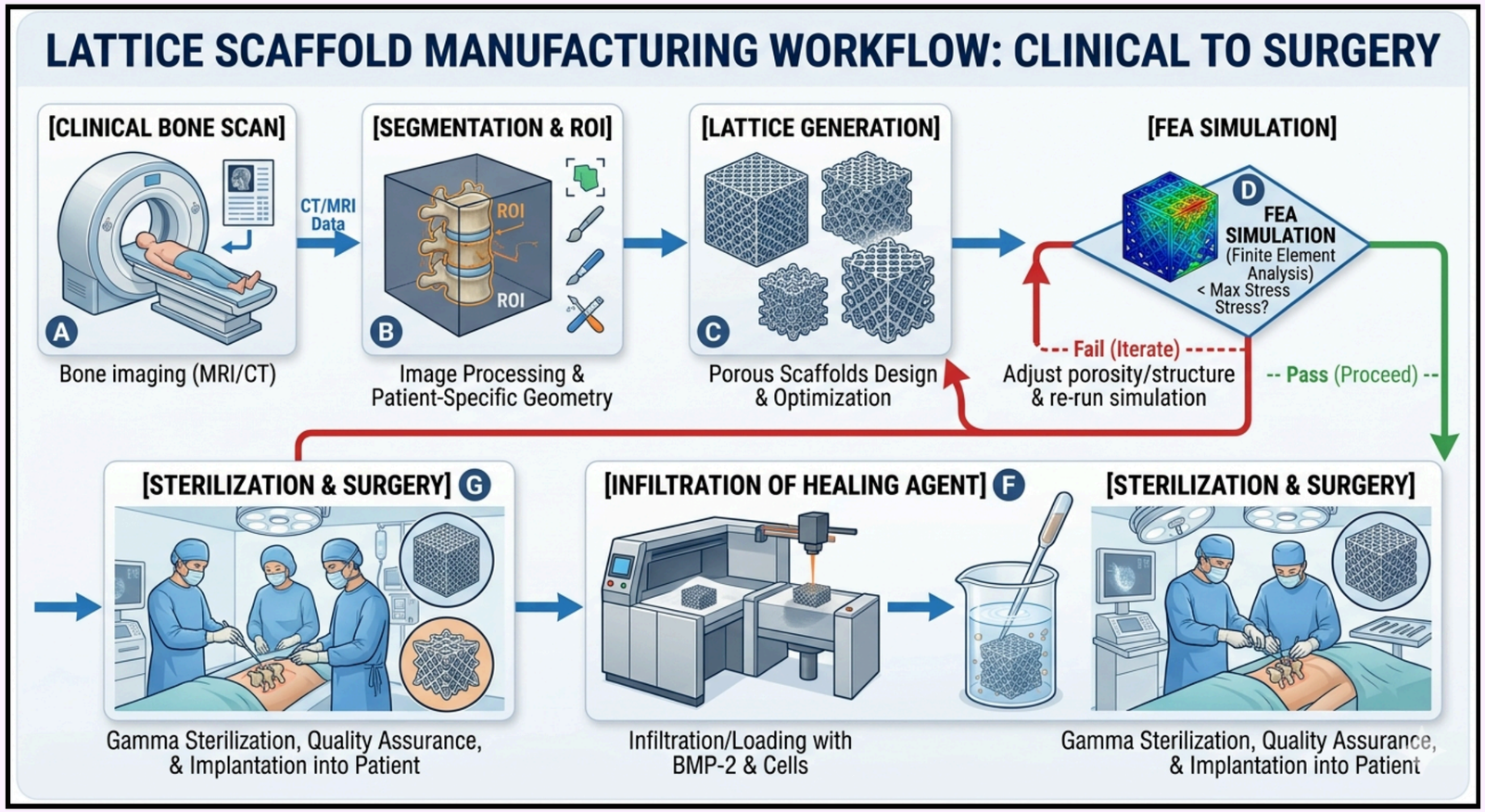
- Modeling Layer: CAD generation of triply periodic minimal surfaces (TPMS).
- Simulation Layer: FEA (Finite Element Analysis) to test load distribution.
- Synthesis Layer: Additive manufacturing and chemical infusion of self-healing monomers.

Design Diagrams

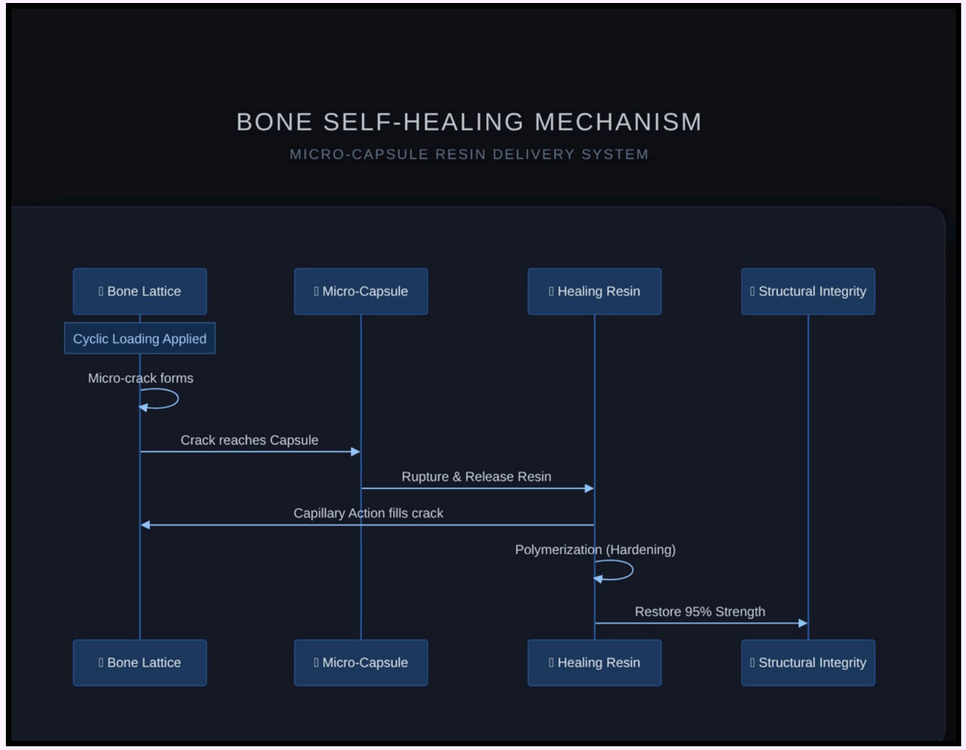
Use Case Diagram:



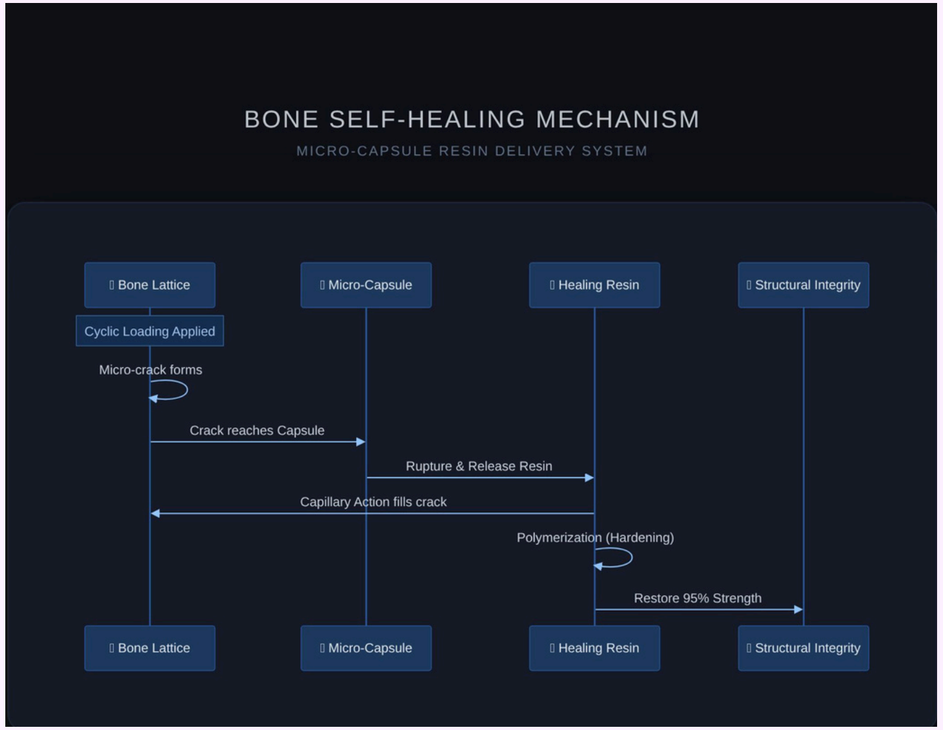
Workflow Diagram:



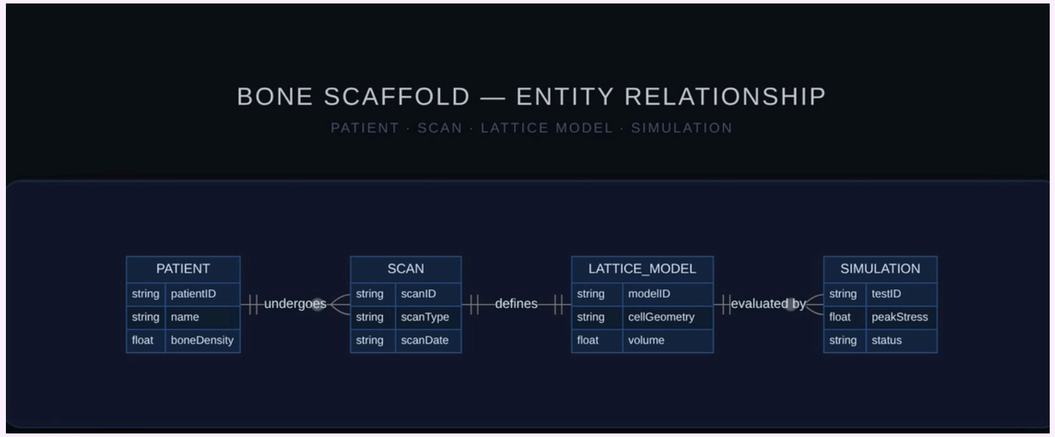
Sequence Diagram:



Class/Component Diagram:



ER Diagram:


BONE SCAFFOLD — ENTITY RELATIONSHIP
PATIENT · SCAN · LATTICE MODEL · SIMULATION
PATIENT
string patientID
string name
float boneDensity
undergoes
SCAN
string scanID
string scanType
string scanDate
defines
LATTICE_MODEL
string modelID
string cellGeometry
float volume
evaluated by
SIMULATION
string testID
float peakStress
string status

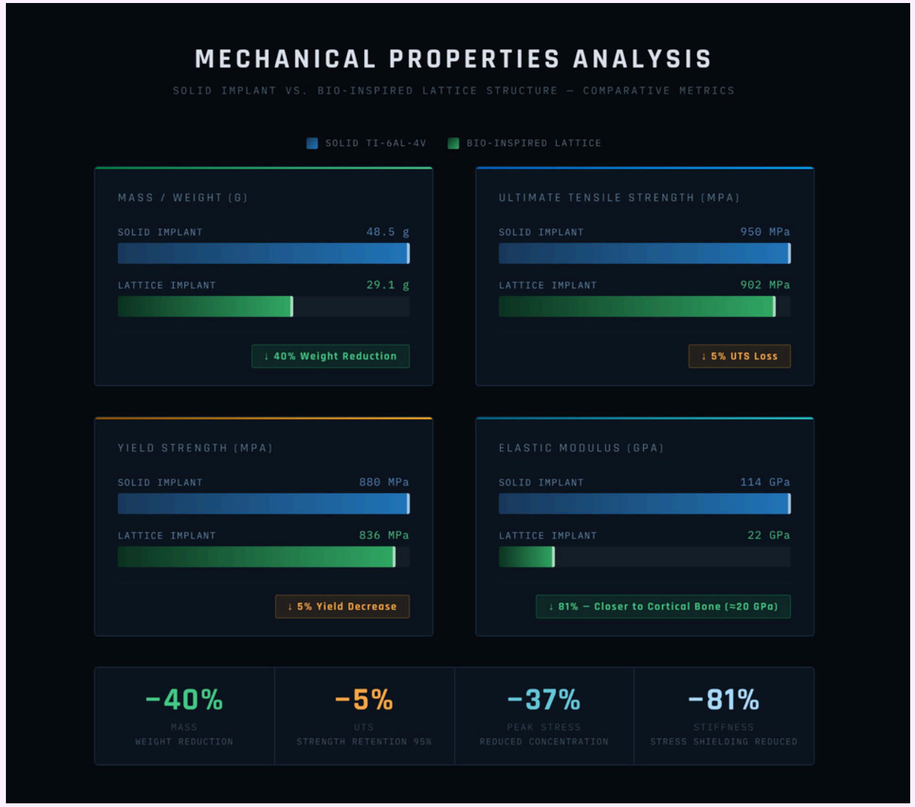
Design Decisions & Rationale:

- Decision: Choosing Gyroid structures over honeycombs.
- Rationale: Gyroids provide better surface-to-volume ratios and more uniform stress distribution, reducing the risk of fracture at the interface.

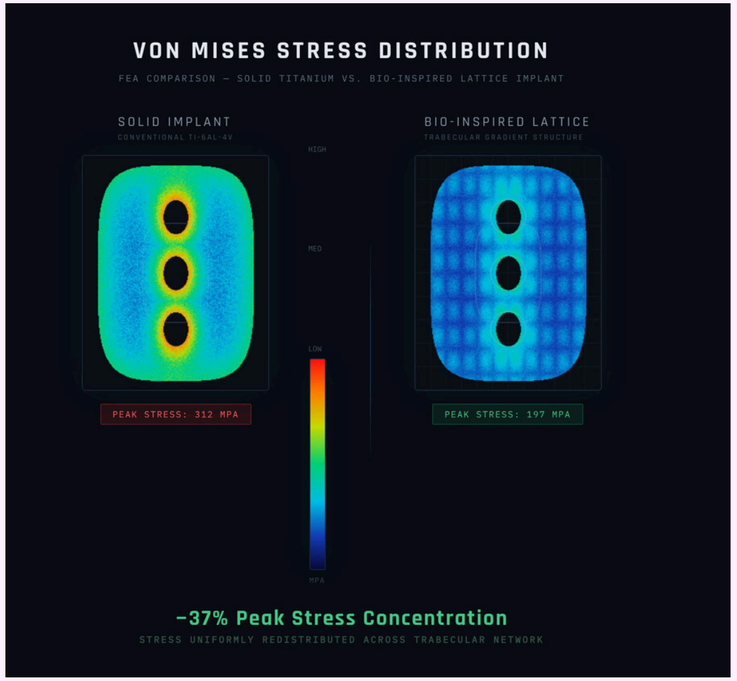
Implementation Details:

- Software: Rhino/Grasshopper for lattice generation; Ansys for structural simulation.
- Material: Ti-6Al-4V alloy infused with a bio-compatible epoxy-based healing agent.

Screenshots / Results:



Mechanical Properties



Stress Heat Maps

Testing Approach:

- Unit Testing: Stress tests on individual lattice cells.
- Integration Testing: Mechanical testing of the lattice-bone interface.
- Self-Healing Test: Deliberate micro-cracking and measurement of strength recovery after 24 hours.

Challenges Faced:

- Complexity in simulating the fluid dynamics of the healing agent within micro-channels.
- Balancing high porosity with the structural demands of the femoral neck.

Learnings & Key Takeaways:

- Deepened understanding of how nature solves structural failure through geometry rather than just material density.
- Mastered the transition from theoretical biomimicry to functional clinical application

Future Enhancements:

- Incorporating 4D printing (materials that change shape based on body temperature).
- Integrating piezoelectric sensors for real-time monitoring of implant health.

References:

1. Gibson, L. J., & Ashby, M. F. Cellular Solids: Structure and Properties.
2. Bhattacharyya, S. (Internal Project Notes, 2026). Bio-Nanotech Applications.
3. Hollister, S. J. (2005). "Porous scaffold design for tissue engineering." Nature Materials, 4(7), 518-524.

THANK YOU!!!